Sahaj Bhadja

(213) 284-4651 sahajbhadja10@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Full-stack Software Engineer with 2+ years building **Java** and Python **REST APIs**, **React** and Next.js interfaces, and access-controlled data models, with automated testing, production support, and eligibility-rule experience.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized multi-tenant data models on Azure SQL with **row-level security** enforcing per-tenant access for 500+ users, and tuned indexed queries to cut dashboard latency from 1.2s to 350ms.
- Built **REST** services and integration pipelines across FastAPI, SQL Server, and PostgreSQL with validation and reconciliation checks, sustaining 99%+ reliability across 5K+ monthly jobs through CI/CD-backed production releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built full-stack features across a **React** dashboard, Flask REST APIs, and MySQL on AWS for upload, rubric, and grading-history flows, serving 150+ instructors and 10K+ daily requests.
- Supported **production reliability** with OCR fallbacks, timeout handling, and multithreaded batch processing that kept APIs stable at peak load, while multimodal AI workflows cut feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built object-oriented **Java** and Python modules with Singleton and Factory patterns for REST API integrations, covered by 20+ Mockito and PyTest **automated tests** across Windows, Linux, and macOS.
- **Debugged** large-scale API integrations through failure logs, then added pagination and field filtering so clients requested only needed data, improving response times 30% and cutting technical debt 20%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **Java Spring Boot** REST microservices and React interfaces for authentication, **role-based access**, customer records, and billing, enforcing JWT-backed permissions while syncing 2M+ records across multiple locations.
- Automated unit, integration, and end-to-end tests with Jest and **Selenium** in Jenkins CI/CD before staged Docker releases, and used Sentry error tracking to resolve issues, reaching 95%+ successful build validation.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built **eligibility rules** for coupons that checked expiry, service restrictions, and usage limits before recomputing prices, inside Flask booking APIs that kept responses under 200ms across 50K+ records.
- Implemented wallet-signature **authentication** for passwordless login and supported a React TypeScript dashboard with Firebase real-time status updates, eliminating stored passwords for wallet users and speeding confirmations 40%.

Skills

Languages: Java, TypeScript, JavaScript, Python, SQL, HTML, CSS
Backend: Spring Boot, REST APIs, Microservices, FastAPI, Flask, Node.js, JSON
Frontend & Practices: React, Next.js, TailwindCSS, React Native, Responsive Design, Data Structures, Agile, Git
Identity & Data: JWT, Role-Based Access Control, Row-Level Security, Authentication, PostgreSQL, Azure SQL, MySQL, Firebase
Quality & Delivery: Jest, Selenium, Mockito, PyTest, Unit Testing, CI/CD, Jenkins, Sentry, Docker, AWS

Projects

Benefits Eligibility Rules Engine | *Java, Spring Boot, JSON Schema, PostgreSQL, Mockito*

- Designed a **Java** service that evaluates JSON-defined eligibility conditions against verified and user-reported attributes, ranks benefit programs for discovery, enforces hard **access gates** per resource, and reports missing attributes.

Accessible Benefits Discovery Portal | *Next.js, React, TypeScript, OAuth/OIDC, Jest, Selenium*

- Designed an **OIDC**-authenticated Next.js portal showing each user's likely benefits with plain-language eligibility reasons, keyboard and screen-reader **accessibility**, and automated Jest and Selenium tests across browsers and edge-case profiles.

HMIS – Hospital Management Information System Live | *MERN, TypeScript, ABDM APIs, Twilio, Sentry*

- Won Smart India Hackathon with a TypeScript healthcare platform using India's ABDM digital-health APIs for **consent-based** patient history access, appointment workflows, and Twilio reminders, cutting administrative workload 30%.